

# 1) Topic

Detecting and analyzing a simulated SSH brute-force attack followed by an unauthorized file transfer (scp/rsync) inside an isolated lab DMZ using tcpdump + host logs.

Simulate attacker (Kali) and victim (Parrot) on isolated VMs, capture traffic, and produce artifacts + a short forensic report.

# 2) Objective

Produce PCAP(s) and host evidence (.log files ) that show:

- multiple failed SSH authentication attempts
- one successful SSH login using a test account
- a follow-up file transfer via scp/rsync (simulated exfil)
- extracted indicators: source IPs, times, login success, file name/size (when possible), and a timeline

*All actions performed only on isolated VMs with test accounts/passwords.*

# 3) Goal (deliverables)

By the end you will have:

1. /tmp/artifacts/ssh\_lab.pcap (full capture) ± transfer\_only.pcap (trimmed)
2. /tmp/artifacts/auth.log (journalctl export) and other host evidence (lastb, last, ss, lsof)
3. Commands & filters used (tcpdump, tshark, grep) documented
4. Timeline (ISO timestamps) and short forensic report with IOCs
5. SHA256 hashes for integrity

## 4) Required / Primary tools (install on each VM as marked)

On Parrot (target):

- **openssh-server** = allows SSH connections
- **tcpdump** = records network traffic
- **tshark** = analyzes captured traffic

On Kali (attacker):

- **openssh-client** → Connects to remote systems and copies files securely (**ssh**, **scp**).
- **sshpass** → Allows using passwords automatically with SSH or SCP.
- **rsync** → Syncs or backs up files efficiently between systems.
- **nmap** → Scans networks to find open ports and services.
- **netcat (nc)** → Sends or receives data over network connections (simple communication tool).

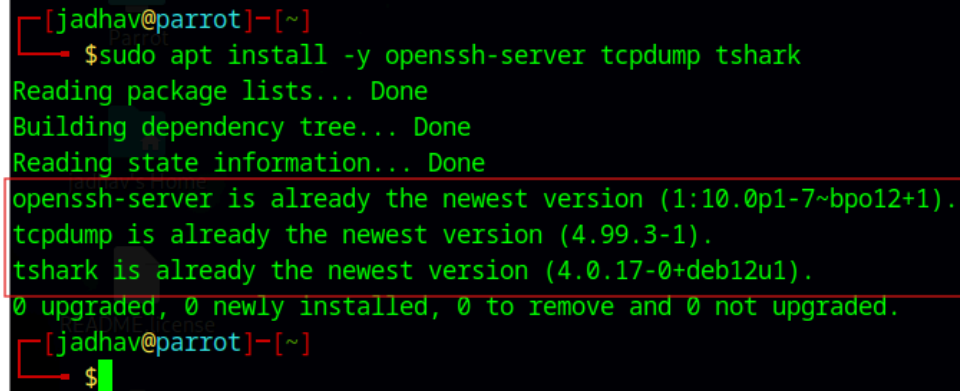
### Install Commands:

# common

sudo apt update

# Parrot (target)

sudo apt install -y openssh-server tcpdump tshark



```
[jadhav@parrot]-[~]  
$ sudo apt install -y openssh-server tcpdump tshark  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
openssh-server is already the newest version (1:10.0p1-7~bpo12+1).  
tcpdump is already the newest version (4.99.3-1).  
tshark is already the newest version (4.0.17-0+deb12u1).  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
[jadhav@parrot]-[~]  
$
```

# Kali (attacker)

sudo apt install -y openssh-client sshpass rsync nmap netcat-openbsd

```
(jadhav@Kali)-[~]
$ sudo apt install -y openssh-client sshpass rsync nmap netcat-openbsd
[sudo] password for jadhav:
openssh-client is already the newest version (1:10.2p1-2).
sshpass is already the newest version (1.10-0.1).
rsync is already the newest version (3.4.1+ds1-6).
rsync set to manually installed.
nmap is already the newest version (7.95+dfsg-3kali1).
```

## Verify Tools:

# Parrot

sudo systemctl enable --now ssh #if this not work use below one

sudo systemctl status ssh --no-pager

```
(jadhav@parrot)-[~]
$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-11 20:46:38 EST; 26min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1045 (sshd)
      Tasks: 1 (limit: 4481)
     Memory: 1.7M
        CPU: 47ms
    CGroup: /system.slice/ssh.service
            └─1045 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 11 20:46:38 parrot systemd[1]: Starting ssh.service - OpenBSD Secure Sh...r...
Nov 11 20:46:38 parrot sshd[1045]: Server listening on 0.0.0.0 port 22.
Nov 11 20:46:38 parrot sshd[1045]: Server listening on :: port 22.
Nov 11 20:46:38 parrot systemd[1]: Started ssh.service - OpenBSD Secure She...ver...
Hint: Some lines were ellipsized, use -l to show in full.
(jadhav@parrot)-[~]
```

ss -tuln | grep ':22' # confirm ssh listening

```
(jadhav@parrot)-[~]
$ ss -tuln | grep ':22' # confirm ssh listening
cp LISTEN 0      128          0.0.0.0:22      0.0.0.0:*
cp LISTEN 0      128          [::]:22       [::]:*
(jadhav@parrot)-[~]
$
```

# tcpdump interfaces

sudo tcpdump -D

```
[jadhav@parrot]~$ sudo tcpdump -D
[sudo] password for jadhav:
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
[jadhav@parrot]~$
```

# tshark version check

tshark -v

```
[jadhav@parrot]~$ tshark -v
TShark (Wireshark) 4.0.17 (Git v4.0.17 packaged as 4.0.17-0+deb12u1).
```

## 5) Preparation (both VMs) — MUST DO BEFORE CAPTURE

1, Check system time manually (optional for timeline alignment):

date -ls

```
(jadhav@Kali)~$ date -ls
2025-11-12T08:00:10+05:30

(jadhav@Kali)~$
```

```
[jadhav@parrot]~$ date -ls
2025-11-12T08:00:59+05:30
[jadhav@parrot]~$
```

## 2, Confirm network mode:

- Use **Bridged** or **Host-only** mode so both VMs can reach each other.
- If NAT: capture from host or bridge interface.

Confirm IPs: ip a

```
(jadhav@Kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:5b:b4:0f brd ff:ff:ff:ff:ff:ff
   inet 172.16.223.193/19 brd 172.16.223.255 scope global dynamic noprefixroute eth0
       valid_lft 256363sec preferred_lft 256363sec
   inet6 fe80::a00:27ff:fe5b:b40f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 1a:44:7e:0f:d5:c6 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

(jadhav@Kali)-[~]

[jadhav@parrot]-[~]
$ ip a | grep -v show timezone file
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:e7:d7:3f brd ff:ff:ff:ff:ff:ff
   inet 172.16.209.85/19 brd 172.16.223.255 scope global dynamic noprefixroute enp0s3
   inet6 fe80::3dd9:3a01:6183:358/64 scope link noprefixroute
       valid_lft 256358sec preferred_lft 256358sec
       valid_lft forever preferred_lft forever
[jadhav@parrot]-[~] $ set-timezone Asia/Kolkata
```

**Observation:** Both VMs are on the **same subnet** (/19 → same network range 172.16.192.0/19). That means they can communicate **without routing** — you just need to confirm ping works.

### 3,Prepare artifacts directory on Parrot:

```
sudo mkdir -p /tmp/artifacts
```

```
[jadhav@parrot]~  
$ sudo mkdir -p /tmp/artifacts  
[sudo] password for jadhav: P> mtu 65536 qdisc  
[jadhav@parrot]~ 00:00:00:00:00 brd 00:00:  
$ 127.0.0.1/8 scope host lo
```

```
sudo chown $USER:$USER /tmp/artifacts
```

```
[jadhav@parrot]~ 00:00:00:00:00 brd 00:00:00:00:00:0  
$ sudo chown $USER:$USER /tmp/artifacts  
[jadhav@parrot]~ ver preferred_lft forever  
$ 6 :1/128 scope host noprefixroute
```

### 4,Configure SSH for lab (Parrot):

```
# enable password authentication & verbose logs
```

```
sudo sed -i 's/^#?PasswordAuthentication.*/PasswordAuthentication yes/' /etc/ssh/sshd_config
```

```
sudo sed -i 's/^#?LogLevel.*/LogLevel VERBOSE/' /etc/ssh/sshd_config
```

```
sudo systemctl restart ssh
```

```
[jadhav@parrot]~  
$ sudo sed -i 's/^#?PasswordAuthentication.*/PasswordAuthentication yes/' /etc/ssh/sshd_c  
onfig: password for jadhav:  
sudo sed -i 's/^#?LogLevel.*/LogLevel VERBOSE/' /etc/ssh/sshd_config  
sudo systemctl restart sshER /tmp/artifacts  
[sudo] password for jadhav:  
[jadhav@parrot]~  
$
```

## 5, Create test user & target file (Parrot):

```
sudo useradd -m -s /bin/bash testuser
echo "testuser:TestPass123!" | sudo chpasswd
echo "lab-secret-data-$(date +%s)" | sudo tee /home/testuser/secret.txt >/dev/null
sudo chown testuser:testuser /home/testuser/secret.txt
stat /home/testuser/secret.txt
```

```
[jadhav@parrot]~$ sudo useradd -m -s /bin/bash testuser
useradd: user 'testuser' already exists
[x]~[jadhav@parrot]~$ echo "testuser:TestPass123!" | sudo chpasswd
[jadhav@parrot]~$ echo "lab-secret-data-$(date +%s)" | sudo tee /home/testuser/secret.txt >/dev/null
[jadhav@parrot]~$ sudo chown testuser:testuser /home/testuser/secret.txt
[jadhav@parrot]~$ stat /home/testuser/secret.txt
  File: /home/testuser/secret.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 0,39    Inode: 16719        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1003/testuser)   Gid: ( 1003/testuser)
Access: 2025-11-07 23:58:07.018875910 +0530
Modify: 2025-11-12 08:13:46.334472311 +0530
Change: 2025-11-12 08:13:55.538458346 +0530
 Birth: 2025-11-07 23:58:07.018875910 +0530
[jadhav@parrot]~$
```

---

## 6) Start Capture (Parrot) — BEFORE ATTACK

### 1,Check interface:

sudo tcpdump -D  
ip a # identify interface (e.g., enp0s3)

```
[jadhav@parrot]~  
→ $sudo tcpdump -D # /PasswordAuthentication: /PasswordAuthentication: yes  
[sudo] password for jadhav:  
1.enp0s3 [Up, Running, Connected] # /VERBOSE: /etc/ssh/ssh_config  
2.any (Pseudo-device that captures on all interfaces) [Up, Running]  
3.lo [Up, Running, Loopback]  
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]  
5.nflog (Linux netfilter log (NFLOG) interface) [none]  
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]  
7.dbus-system (D-Bus system bus) [none]  
8.dbus-session (D-Bus session bus) [none] dbpasswd  
[jadhav@parrot]~  
→ $
```

### 2,Start tcpdump (SSH only):

sudo tcpdump -i enp0s3 -nn -tttt port 22 -w /tmp/artifacts/ssh\_lab.pcap &  
echo \$! > /tmp/artifacts/tcpdump.pid

```
[jadhav@parrot]~  
→ $ sudo tcpdump -i enp0s3 -nn -tttt port 22 -w /tmp/artifacts/ssh_lab.pcap &  
echo $! > /tmp/artifacts/tcpdump.pid  
[1] 5254  
[jadhav@parrot]~  
→ $tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
$
```

### 3,Optional — full backup capture:

sudo tcpdump -i enp0s3 -nn -tttt -w /tmp/artifacts/all\_lab.pcap &

```
[jadhav@parrot]~  
→ $sudo tcpdump -i enp0s3 -nn -tttt -w /tmp/artifacts/all_lab.pcap &  
[1] 5343  
[jadhav@parrot]~  
→ $
```

### 4,Monitor SSH logs live:

sudo journalctl -u ssh -f --no-pager &

```
[jadhav@parrot]~  
→ $ sudo journalctl -u ssh -f --no-pager &  
[1] 5384  
[jadhav@parrot]~  
→ $
```



# 7) Attacker Simulation (Kali) — AFTER CAPTURE STARTED

1, Check connectivity & open port:

```
ping -c 3 172.16.209.85
```

```
(jadhav@Kali)-[~]
$ ping -c 3 172.16.209.85
PING 172.16.209.85 (172.16.209.85) 56(84) bytes of data:
64 bytes from 172.16.209.85: icmp_seq=1 ttl=64 time=5.11 ms
64 bytes from 172.16.209.85: icmp_seq=2 ttl=64 time=1.61 ms
64 bytes from 172.16.209.85: icmp_seq=3 ttl=64 time=1.70 ms

— 172.16.209.85 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.614/2.808/5.111/1.628 ms

(jadhav@Kali)-[~]
$
```

```
nmap -Pn -p 22 172.16.209.85
```

```
(jadhav@Kali)-[~]
$ nmap -Pn -p 22 172.16.209.85
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-12 08:22 IST
Nmap scan report for 172.16.209.85
Host is up (0.0029s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:E7:D7:3F (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds

(jadhav@Kali)-[~]
$
```

2, Simulate failed SSH attempts:

**sshpas** is a Linux tool that lets you automatically provide a password to SSH commands

```
for i in {1..3}; do
  date -ls
  sshpass -p "WrongPass$i" ssh -o StrictHostKeyChecking=no -o ConnectTimeout=5
  testuser@172.16.209.85 "echo try-$i; exit"
done
```

```
(jadhav@Kali)-[~]
$ for i in {1..3}; do
  date -Is
  sshpass -p "WrongPass$i" ssh -o StrictHostKeyChecking=no -o ConnectTimeout=5 testuser@172.16.209.85 "echo try-$i; e
xit"
done
2025-11-12T08:24:37+05:30
Permission denied, please try again.
2025-11-12T08:24:39+05:30
Permission denied, please try again.
2025-11-12T08:24:42+05:30
Permission denied, please try again.
```

### 3,Successful login:

```
date -Is
sshpass -p "TestPass123!" ssh -o StrictHostKeyChecking=no testuser@172.16.209.85 'whoami;
hostname; exit'
```

```
(jadhav@Kali)-[~]
$ date -Is
sshpass -p "TestPass123!" ssh -o StrictHostKeyChecking=no testuser@172.16.209.85 'whoami; hostname; exit'
2025-11-12T08:25:52+05:30
testuser
parrot
```

### 4,Simulate exfiltration (scp or rsync):

**scp** → stands for *secure copy*.

- ♦ Used to copy files between your local system and a remote system over SSH.

```
# scp pull (attacker steals file)
date -Is
sshpass -p "TestPass123!" scp -o StrictHostKeyChecking=no
testuser@172.16.209.85:/home/testuser/secret.txt ./stolen_secret.txt
```

```
(jadhav@Kali)-[~]
$ date -Is
sshpass -p "TestPass123!" scp -o StrictHostKeyChecking=no testuser@172.16.209.85:/home/testuser/secret.txt ./stolen_s
ecret.txt
2025-11-12T08:27:06+05:30
```

# OR rsync alternative

**rsync** → stands for *remote sync*.

- ♦ Used to **sync files or folders**, only transferring the changes — faster than scp for repeated transfers.

```
mkdir -p ~/exfil_out
sshpass -p "TestPass123!" rsync -av -e "ssh -o StrictHostKeyChecking=no"
testuser@172.16.209.85:/home/testuser/secret.txt ~/exfil_out/
```

```
(jadhav@Kali)-[~]
$ mkdir -p ~/exfil_out
sshpas -p "TestPass123!" rsync -av -e "ssh -o StrictHostKeyChecking=no" testuser@172.16.209.85:/home/testuser/secret
.txt ~/exfil_out/
receiving incremental file list
secret.txt

sent 49 bytes  received 144 bytes  128.67 bytes/sec
total size is 27  speedup is 0.14

(jadhav@Kali)-[~]
$
```

## 8) Stop Capture & Collect Evidence (Parrot)

### 1,Stop tcpdump safely:

```
sudo kill -INT "$(cat /tmp/artifacts/tcpdump.pid)" || sudo pkill -f "tcpdump -i enp0s3"
ls -lh /tmp/artifacts/ssh_lab.pcap
```

```
(jadhav@parrot)-[~]
$ sudo kill -INT "$(cat /tmp/artifacts/tcpdump.pid)" || sudo pkill -f "tcpdump -i enp0s3"
ls -lh /tmp/artifacts/ssh_lab.pcap
[sudo] password for jadhav:
-rw-r--r-- 1 tcpdump tcpdump 128K Nov 12 08:27 /tmp/artifacts/ssh_lab.pcap
(jadhav@parrot)-[~]
$
```

### 2,Export SSH logs and system evidence:

```
sudo journalctl -u ssh --no-pager --since "1 hour ago" > /tmp/artifacts/auth.log
sudo journalctl -u ssh --no-pager > /tmp/artifacts/auth_full.log
sudo cp /var/log/auth.log /tmp/artifacts/ 2>/dev/null || true
sudo last -n 20 > /tmp/artifacts/last.log
sudo lastb -n 50 > /tmp/artifacts/lastb.log || true
sudo ss -tnp | grep ':22' > /tmp/artifacts/ss_22.txt || true
sudo lsof -iTCP:22 -sTCP:ESTABLISHED -P -n > /tmp/artifacts/lsof_ssh.txt || true
stat /home/testuser/secret.txt > /tmp/artifacts/secret_stat.txt
ls -l --full-time /home/testuser/secret.txt >> /tmp/artifacts/secret_stat.txt
```

```

[jadhav@parrot]~$ sudo journalctl -u ssh --no-pager --since "1 hour ago" > /tmp/artifacts/auth.log
sudo journalctl -u ssh --no-pager > /tmp/artifacts/auth_full.log
sudo cp /var/log/auth.log /tmp/artifacts/ &>/dev/null || true
sudo last -n 20 > /tmp/artifacts/last.log
sudo lastb -n 50 > /tmp/artifacts/lastb.log || true
sudo ss -tnp | grep ':22' > /tmp/artifacts/ss_22.txt || true
sudo lsof -iTCP:22 -sTCP:ESTABLISHED -P -n > /tmp/artifacts/lsof_ssh.txt || true
stat /home/testuser/secret.txt > /tmp/artifacts/secret_stat.txt
ls -l --full-time /home/testuser/secret.txt >> /tmp/artifacts/secret_stat.txt
Journal file /var/log/journal/f5fb62995b6249b58e3fd6fbd10def7d/user-1000@000642aa166a1263-240dff3daaf670af.journal~ is truncated, ignoring file.
Journal file /var/log/journal/f5fb62995b6249b58e3fd6fbd10def7d/user-1000@000642aa166a1263-240dff3daaf670af.journal~ is truncated, ignoring file.
[jadhav@parrot]~$

```

### 3Compute hashes:

sha256sum /tmp/artifacts/ssh\_lab.pcap > /tmp/artifacts/ssh\_lab.pcap.sha256

sha256sum /tmp/artifacts/auth.log > /tmp/artifacts/auth.log.sha256

sudo chown \$USER:\$USER /tmp/artifacts/\* || true

ls -l /tmp/artifacts

```

[jadhav@parrot]~$
$ sha256sum /tmp/artifacts/ssh_lab.pcap > /tmp/artifacts/ssh_lab.pcap.sha256
sha256sum /tmp/artifacts/auth.log > /tmp/artifacts/auth.log.sha256
sudo chown $USER:$USER /tmp/artifacts/* || true
ls -l /tmp/artifacts jadhav:
[sudo] password for jadhav: p128K Nov 12 08:27 /tmp/artifacts/ssh_lab.pcap
total 1756 parrot]~$
-rw-r--r-- 1 jadhav jadhav 1619986 Nov 12 08:30 auth_full.log > /tmp/artifacts/auth.log
-rw-r--r-- 1 jadhav jadhav 13234 Nov 12 08:30 auth.log full.log
-rw-r--r-- 1 jadhav jadhav 90 Nov 12 08:32 auth.log.sha256
-rw-r--r-- 1 jadhav jadhav 3688 Nov 12 08:30 lastb.log
-rw-r--r-- 1 jadhav jadhav 1493 Nov 12 08:30 last.log
-rw-r--r-- 1 jadhav jadhav 0 Nov 12 08:30 lsof_ssh.txt
-rw-r--r-- 1 jadhav jadhav 488 Nov 12 08:30 secret_stat.txt f_ssh.txt || true
-rw-r--r-- 1 jadhav jadhav 0 Nov 12 08:30 ss_22.txt stat.txt
-rw-r--r-- 1 jadhav jadhav 131212 Nov 12 08:29 ssh_lab.pcaps/secret_stat.txt
-rw-r--r-- 1 jadhav jadhav 94 Nov 12 08:32 ssh_lab.pcap.sha256 1000@000642aa166a
-rw-r--r-- 1 jadhav jadhav 5 Nov 12 08:17 tcpdump.pid
[jadhav@parrot]~$ /journal/f5fb62995b6249b58e3fd6fbd10def7d/user-1000@000642aa166a
$ 670af.journal~ is truncated, ignoring file.

```

## 9) Packet & Log Analysis(Run on Parrot )

### 1, TCP Conversations (find large flow):

```
tshark -r /tmp/artifacts/ssh_lab.pcap -q -z conv,tcp
```

```
[jadhav@parrot:~]$ tshark -r /tmp/artifacts/ssh_lab.pcap -q -z conv,tcp /tmp/artifacts/ssh_lab.pcap:10240
=====
TCP Conversations: /tmp/artifacts/ssh_lab.pcap
Filter: <No Filter>

  172.16.223.193:39132 <--> 172.16.209.85:22 32 6,561 bytes 30 6,129 bytes 62 12 kB 309.892749000 1.4861
  172.16.223.193:43014 <--> 172.16.209.85:22 28 6,569 bytes 28 5,993 bytes 56 12 kB 260.751248000 1.4696
  172.16.223.193:35304 <--> 172.16.209.85:22 22 5,477 bytes 22 5,161 bytes 44 10 kB 186.650850000 1.6198
  172.16.223.193:54520 <--> 172.16.209.85:22 13 3,887 bytes 15 4,175 bytes 28 8,062 bytes 45.525859000 3.9002
  172.16.223.193:32854 <--> 172.16.209.85:22 14 3,965 bytes 14 4,229 bytes 28 8,194 bytes 53.255649000 3.7641
  172.16.223.193:59914 <--> 172.16.209.85:22 13 3,899 bytes 14 4,229 bytes 27 8,128 bytes 72.150506000 3.7677
  172.16.223.193:54524 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 49.432776000 3.8161
  172.16.223.193:32866 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 57.052742000 3.7776
  172.16.223.193:57680 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 60.840555000 3.7661
  172.16.223.193:57616 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 64.611776000 3.7545
  172.16.223.193:57626 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 68.387342000 3.7525
  172.16.223.193:42668 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 111.683178000 1.9788
  172.16.223.193:42680 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 113.668553000 3.8045
  172.16.223.193:42684 <--> 172.16.209.85:22 13 3,887 bytes 13 4,043 bytes 26 7,930 bytes 117.485778000 3.7929
  172.16.223.193:33790 <--> 172.16.209.85:22 1 58 bytes 2 120 bytes 3 178 bytes 0.000000000 0.0061
=====
[jadhav@parrot:~]$
```

## 2,Show SSH packets (timestamp/src/dst):

```
tshark -r /tmp/artifacts/ssh_lab.pcap -Y "tcp.port==22" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.srcport -e tcp.dstport -e frame.len
```

```

Parrot Terminal
File Edit View Search Terminal Help
[jadhav@parrot]~$ 
[jadhav@parrot]~$ tshark -r /tmp/artifacts/ssh_lab.pcap -Y "tcp.port==22" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.srcport -e tcp.dstport -e frame.len
100 Nov 12, 2025 08:22:45.394633000 IST 172.16.223.193 172.16.209.85 33790 22 60
211 Nov 12, 2025 08:22:45.396846000 IST 172.16.209.85 172.16.223.193 22 3379058
3 Nov 12, 2025 08:22:45.400691000 IST 172.16.223.193 172.16.209.85 33790 22 60
4 Nov 12, 2025 08:23:30.920492000 IST 172.16.223.193 172.16.209.85 54520 22 74
572 Nov 12, 2025 08:23:30.921027000 IST 172.16.209.85 172.16.223.193 22 5452074
672 Nov 12, 2025 08:23:30.924032000 IST 172.16.223.193 172.16.209.85 54520 22 66
772 Nov 12, 2025 08:23:30.928664000 IST 172.16.223.193 172.16.209.85 54520 22 99
872 Nov 12, 2025 08:23:30.929720000 IST 172.16.209.85 172.16.223.193 22 5452066
972 Nov 12, 2025 08:23:31.026705000 IST 172.16.209.85 172.16.223.193 22 54520107
1072 Nov 12, 2025 08:23:31.029141000 IST 172.16.223.193 172.16.209.85 54520 22 66
1172 Nov 12, 2025 08:23:31.045819000 IST 172.16.223.193 172.16.209.85 54520 22 1634
1272 Nov 12, 2025 08:23:31.045851000 IST 172.16.209.85 172.16.223.193 22 5452066
1372 Nov 12, 2025 08:23:31.078237000 IST 172.16.209.85 172.16.223.193 22 545201106
1472 Nov 12, 2025 08:23:31.082088000 IST 172.16.223.193 172.16.209.85 54520 22 1298
1572 Nov 12, 2025 08:23:31.106873000 IST 172.16.209.85 172.16.223.193 22 545201646
1672 Nov 12, 2025 08:23:31.108212000 IST 172.16.223.193 172.16.209.85 54520 22 66
1772 Nov 12, 2025 08:23:31.115509000 IST 172.16.223.193 172.16.209.85 54520 22 150
1872 Nov 12, 2025 08:23:31.158204000 IST 172.16.209.85 172.16.223.193 22 5452066
1972 Nov 12, 2025 08:23:31.160339000 IST 172.16.223.193 172.16.209.85 54520 22 110
2072 Nov 12, 2025 08:23:31.160501000 IST 172.16.209.85 172.16.223.193 22 5452066
2172 Nov 12, 2025 08:23:31.160742000 IST 172.16.209.85 172.16.223.193 22 54520110
2272 Nov 12, 2025 08:23:31.163151000 IST 172.16.223.193 172.16.209.85 54520 22 134
2372 Nov 12, 2025 08:23:31.168134000 IST 172.16.209.85 172.16.223.193 22 54520330
2472 Nov 12, 2025 08:23:31.212482000 IST 172.16.223.193 172.16.209.85 54520 22 66
2572 Nov 12, 2025 08:23:31.228264000 IST 172.16.223.193 172.16.209.85 54520 22 214
2672 Nov 12, 2025 08:23:31.324228000 IST 172.16.209.85 172.16.223.193 22 5452066
2772 Nov 12, 2025 08:23:34.814052000 IST 172.16.209.85 172.16.223.193 22 54520118
2872 Nov 12, 2025 08:23:34.815202000 IST 172.16.223.193 172.16.209.85 54520 22 66
2972 Nov 12, 2025 08:23:34.816344000 IST 172.16.223.193 172.16.209.85 54520 22 66

```

### 3,SYN/FIN/RST events (failed attempts):

```

tshark -r /tmp/artifacts/ssh_lab.pcap -Y "tcp.flags.syn==1 || tcp.flags.reset==1 || tcp.flags.fin==1" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.flags

```

```
Parrot Terminal
File Edit View Search Terminal Help
[jadhav@parrot]~$
[jadhav@parrot]~$ tshark -r /tmp/artifacts/ssh_lab.pcap -Y "tcp.flags.syn==1 || tcp.flags.reset==1 || tcp.flags.fin==1" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.flags
1      Nov 12, 2025 08:22:45.394633000 IST      172.16.223.193 172.16.209.85 0x0002 22 60
2      Nov 12, 2025 08:22:45.396846000 IST      172.16.209.85 172.16.223.193 0x0012 3370058
3      Nov 12, 2025 08:22:45.400691000 IST      172.16.223.193 172.16.209.85 0x0004 22 60
4      Nov 12, 2025 08:23:30.920492000 IST      172.16.223.193 172.16.209.85 0x0002 22 74
5      Nov 12, 2025 08:23:30.921027000 IST      172.16.209.85 172.16.223.193 0x0012 5452078
29     Nov 12, 2025 08:23:34.816344000 IST      172.16.223.193 172.16.209.85 0x0011 22 66
30     Nov 12, 2025 08:23:34.819821000 IST      172.16.209.85 172.16.223.193 0x0011 22 98
32     Nov 12, 2025 08:23:34.827409000 IST      172.16.223.193 172.16.209.85 0x0002 5452066
33     Nov 12, 2025 08:23:34.827576000 IST      172.16.209.85 172.16.223.193 0x0012 54520187
55     Nov 12, 2025 08:23:38.624887000 IST      172.16.223.193 172.16.209.85 0x0011 22 66
56     Nov 12, 2025 08:23:38.636761000 IST      172.16.209.85 172.16.223.193 0x0011 22 1634
58     Nov 12, 2025 08:23:38.650282000 IST      172.16.223.193 172.16.209.85 0x0002 5452066
59     Nov 12, 2025 08:23:38.650455000 IST      172.16.209.85 172.16.223.193 0x0012 545201186
83     Nov 12, 2025 08:23:42.405358000 IST      172.16.223.193 172.16.209.85 0x0011 22 1298
84     Nov 12, 2025 08:23:42.412451000 IST      172.16.209.85 172.16.223.193 0x0011 545201646
86     Nov 12, 2025 08:23:42.447375000 IST      172.16.223.193 172.16.209.85 0x0002 22 66
87     Nov 12, 2025 08:23:42.447917000 IST      172.16.209.85 172.16.223.193 0x0012 22 150
109    Nov 12, 2025 08:23:46.218472000 IST      172.16.223.193 172.16.209.85 0x0011 5452066
110    Nov 12, 2025 08:23:46.223724000 IST      172.16.209.85 172.16.223.193 0x0011 22 110
112    Nov 12, 2025 08:23:46.235188000 IST      172.16.223.193 172.16.209.85 0x0002 5452066
113    Nov 12, 2025 08:23:46.235328000 IST      172.16.209.85 172.16.223.193 0x0012 54520118
135    Nov 12, 2025 08:23:49.995879000 IST      172.16.223.193 172.16.209.85 0x0011 22 134
136    Nov 12, 2025 08:23:50.000171000 IST      172.16.209.85 172.16.223.193 0x0011 54520330
138    Nov 12, 2025 08:23:50.012399000 IST      172.16.223.193 172.16.209.85 0x0002 22 66
139    Nov 12, 2025 08:23:50.012518000 IST      172.16.209.85 172.16.223.193 0x0012 22 214
161    Nov 12, 2025 08:23:53.760902000 IST      172.16.223.193 172.16.209.85 0x0011 5452066
```

#### 4,Extract failed/successful login lines:

```
sudo journalctl -u ssh --no-pager --output=short-iso | egrep -i "Failed password|Accepted password" > /tmp/artifacts/auth_filtered_iso.log
```

```
File Edit View Search Terminal Help
[jadhav@parrot]~$
[jadhav@parrot]~$ sudo journalctl -u ssh --no-pager --output=short-iso | egrep -i "Failed password|Accepted password" > /tmp/artifacts/auth_filtered_iso.log
[sudo] password for jadhav:
Journal file /var/log/journal/f5fb62995b6249b58e3fd6fdb10def7d/user-1000@000642aa166a1263-240dff3daaf670af.journal- is truncated, ignoring file.
[jadhav@parrot]~$
```